Uncertainty Visualization for Biomolecular Structures: An Empirical Evaluation

Anna Sterzik, Michael Krone, *Member, IEEE*, Daniel Baum, Douglas W. Cunningham, and Kai Lawonn

***Abstract*—Uncertainty is an intrinsic property of almost all data, regardless of the data being measured, simulated, or generated. It can significantly influence the results and reliability of subsequent analysis steps. Clearly communicating uncertainties is crucial for informed decision-making and understanding, especially in biomolecular data, where uncertainty is often difficult to infer. Uncertainty visualization (UV) is a powerful tool for this purpose. However, previously proposed uncertainty visualization (UV) methods lack sufficient empirical evaluation. We collected and categorized visualization methods for portraying positional uncertainty in biomolecular structures. We then organized the methods into metaphorical groups and extracted nine representatives: color, clouds, ensemble, hulls, sausages, contours, texture, waves, and noise. We assessed their strengths and weaknesses in a twofold approach: expert assessments with six domain experts and three perceptual evaluations involving 1,756 participants. Through the expert assessments, we aimed to highlight the advantages and limitations of the individual methods for the application domain and discussed areas for necessary improvements. Through the perceptual evaluation, we investigated whether the visualizations are intuitively associated with uncertainty and whether the directionality of the mapping is perceived as intended. We also assessed the accuracy of inferring uncertainty values from the visualizations. Based on our results, we judged the appropriateness of the metaphors for encoding uncertainty and suggest further areas for improvement.**

***Index Terms*—Biomolecular visualization, perceptual evaluation, uncertainty, flexibility, ensemble visualization.**

I. Introduction

MOLECULAR visualization, with its long and rich history, has seen the development of various methods to understand the structure and function of molecules [1]. Although there have been many significant advancements in biomolecular visualization, one crucial but often neglected aspect is the visualization of uncertainty. While the importance of visualizing uncertainty has been emphasized [2] and even classified as one of the top scientific visualization research problems 20 years ago [3], UV remains largely absent in biomolecular contexts. This absence risks implying certainty where none exists, potentially resulting in erroneous assessments. Even though it is still relatively uncommon to visualize uncertainty in biomolecular contexts, several approaches—such as ensemble visualization, color mapping, or uncertainty hulls—have surfaced. Empirical evaluations of these techniques are still scarce. The lack of comprehensive empirical studies comparing different types of biomolecular UV was recently identified as one of the most important open problems in uncertainty-aware visualization of biomolecular structures [4].

To address this gap, we investigated various methods for representing positional uncertainty in proteins. Our methodology involved two main strategies: expert assessment and crowdsourced studies. Domain experts provided detailed insights into the strengths and weaknesses of each metaphor in the context of biomolecular structure visualization and highlighted further areas for improvement. The crowdsourced studies aimed at understanding the perception of the visualizations. Initially, we focused on the spontaneous associations evoked by the visualizations. We determined whether viewers naturally understand that the representations depict uncertainty without any explicit prompting or instruction. In the second study, we investigated whether the directionality of the metaphor used to depict uncertainty aligns with how participants perceive uncertainty. More specifically, we investigated whether areas characterized by high uncertainty are correctly identified as such or are misconceived as low-uncertainty areas. In the final study, we evaluated the accuracy and ease of judgments regarding uncertainty within specific regions. Fig. 1 shows a graphical summary of our evaluation workflow. For examples of the uncertainty metaphors, please refer to Fig. 2.

We determined some interesting discrepancies between the expert judgments and the perception of the metaphors. The experts favored familiar representations and gave mixed feedback for less conventional metaphors. However, in our perceptual evaluations, some of the less established metaphors performed very well, while more common metaphors led to worse results. Our contributions can be summarized as follows: we provide novel insights into the domain-specific utility, intuitiveness, and understandability of nine visual metaphors to convey uncertainty

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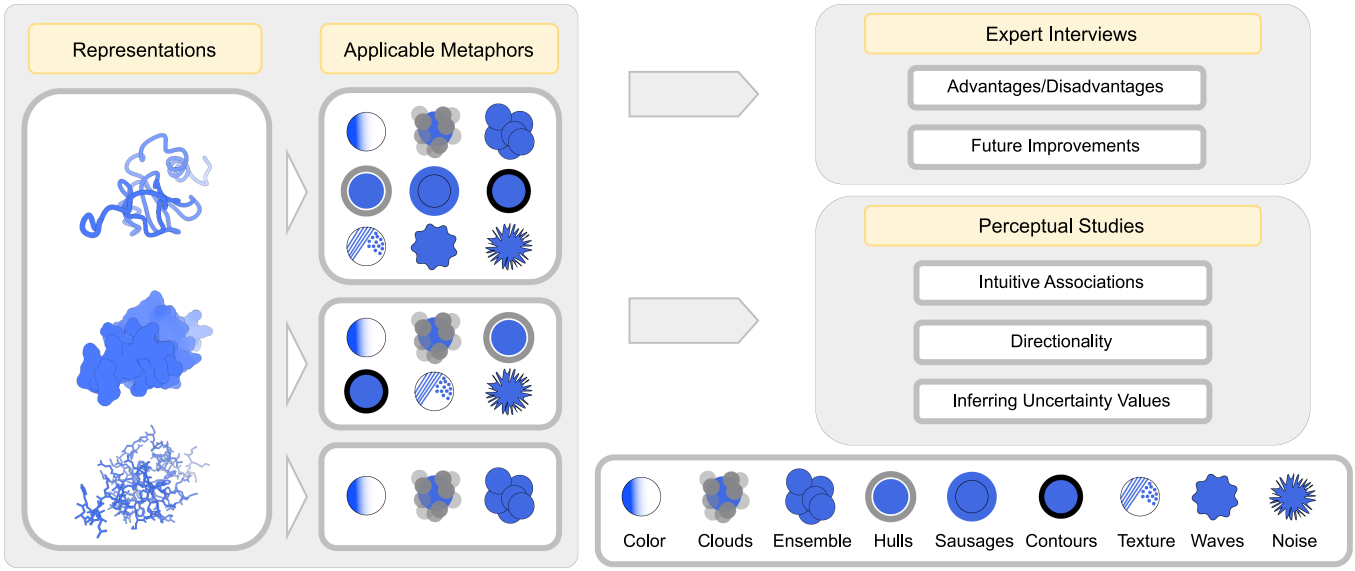


Fig. 1. Our evaluation workflow for biomolecular uncertainty visualization tests nine metaphors (bottom right) across three protein representations (left, from top to bottom: tube, SES, sticks). Certain metaphors are tailored to specific representations. We integrate expert interviews with three perceptual studies.

in molecular structures. Our results are based on domain expert assessments and large-scale perceptual studies.

II. Background

In the following, we discuss sources of biomolecular uncertainty and typical visualization approaches.

A. Uncertainty in Biomolecules

Molecules—especially biomolecules, the nanoscopic building blocks of all life on earth—have been studied extensively. Among biomolecules, proteins play a crucial role in virtually all biological processes. Proteins are macromolecules that consist of one or more chains of amino acids. Each amino acid has an identical so-called backbone part and an individual small side chain of up to 20 atoms that constitutes the physicochemical properties of this amino acid. The protein's amino acid chains fold into semi-stable conformations, or secondary structures, that are stabilized mainly by hydrogen bonds. The three-dimensional arrangement of these secondary structures is called tertiary structure and defines the protein's shape. Finally, the quaternary structure is an assembly of multiple chains forming a functional compound.

The three-dimensional structure of a protein is crucial for its function. Misfolded proteins can cause severe illnesses [5]. However, despite proteins forming tertiary structures, they still have a lot of flexibility that enables parts of the protein chain to move to fulfill certain functions, e.g., a protein forming a transmembrane channel that opens and closes [6]. While the inner core of most proteins is usually rather rigid, the outer parts can be more flexible. This flexibility of the protein can make it more unlikely for other small molecules to bind to the surface of the protein. Therefore, bioengineering often tries to stabilize proteins by introducing small mutations, i.e., swapping one amino acid with another one that has similar properties. This change should not alter the function of the protein but only make it less flexible. However, there also exist *intrinsically disordered proteins* (IDP), which have no fixed structure [7]. Some of these IDPs fold into a stable structure when being in contact with a certain reaction partner or another specific molecule, but some IDPs can even bind to other molecules and execute their function without folding into a fixed configuration.

In summary, proteins, and biomolecules in general, are subject to a lot of structural flexibility. This flexibility leads to what is commonly referred to as *positional uncertainty* [8], [9]. Since the structure of a molecule influences its function, this uncertainty can have a great effect on the molecule's function.

Different data sources provide distinct measures of uncertainty. For instance, data from X-ray crystallography typically include an uncertainty metric known as the b-factor. This measure reflects the attenuation of X-ray scattering caused by the thermal motion of atoms. Another common data source is molecular ensembles, which consist of multiple conformations of the same molecule. These ensembles can be derived experimentally, such as through nuclear magnetic resonance (NMR) spectroscopy, which captures molecules in solution and reflects their inherent flexibility and dynamic behavior. Ensembles can also be generated computationally using molecular dynamics (MD) simulations, which model molecular motions over time and explore the range of possible conformations a molecule can adopt. To quantify uncertainty within ensembles, scalar measures such as the root mean square fluctuation (RMSF) are often used. RMSF describes the average positional deviation of individual atoms over time or across conformations, providing a numerical expression of uncertainty.

B. Molecular Visualization

From the very beginning of using computers to study molecular structure, properties, and function, visualization has played an important role. Over time, the early approaches in molecular graphics—that is, the graphical depiction of molecular models

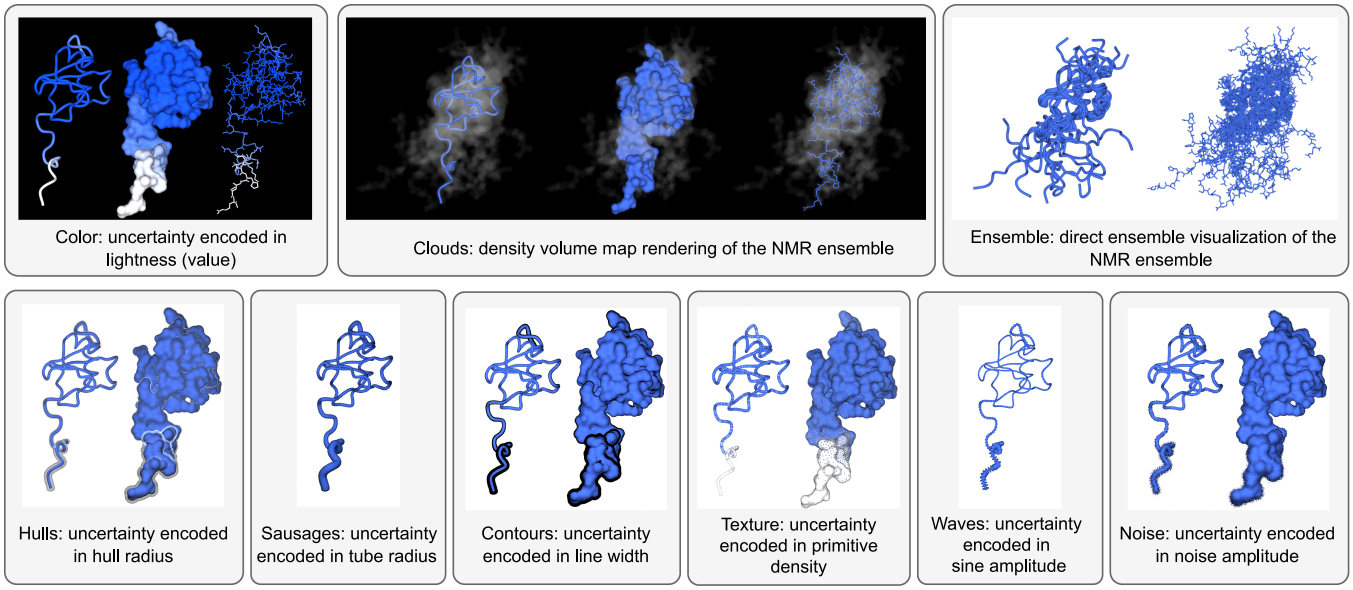


Fig. 2. Overview of the nine representative UV metaphors used in our evaluation.

ranging back more than five decades—have evolved into modern, feature-rich visual analytics tools. To show different aspects of a molecule's structure, various representations have been developed, ranging from simple ball-and-stick or space-filling models to complex molecular surfaces or abstract representations for specific macromolecules like proteins or DNA.

According to the state-of-the-art report by Kozlíková et al. [1], the most important representation models for biomolecular structures can be divided into *atomistic models* and *abstract models*. As the name implies, atomistic models take into account all atoms of a biomolecule. These representations include the well-known *ball-and-stick* model, where each atom is represented by a small sphere, and bonds are visualized as thin cylinders connecting the respective atom spheres. The *stick* model (sometimes also called *licorice*) is a variation of this: here, the cylinders representing the bonds are typically a bit thicker, and the atom spheres are rendered with the same radius, so they only appear as rounded caps of the cylinders.

Other types of atomistic models are surface models that depict the interface between a molecule and its environment. The simple and often used *van der Waals* (vdW) surface shows each atom as a sphere with the vdW radius of the corresponding element. The vdW radius corresponds to the distance of non-bonded atoms, that is, the vdW surface is an approximate depiction of the interaction surface with another molecule. The downside of the vdW surface is that it often has many small clefts and crevices and that it does not show the interaction surface with respect to a specific molecule. Thus, the *solvent excluded surface* (SES) is often preferred: it shows the surface is accessible to a small molecule of a certain size, modeled by a spherical probe. This probe is rolled over the vdW surface and traces out the SES.

In contrast to atomistic models, abstract models do not show the individual atoms of the molecule. A common abstraction for proteins is the *backbone* or *tube* model that uses a central carbon atom of the backbone of each amino acid as control points for a spline. The spline is often rendered as a tube to make it more visible. This tube visualizes the shape of the backbone and, thus, the structure of the amino acid chain. A similar abstraction is commonly used for DNA, creating the well-known twisted ladder appearance of the DNA double helix.

III. Related Work and Categorization of Uncertainty Metaphors

For our evaluation, we collected methods that visualize positional uncertainty in biomolecules. We focused on techniques that directly alter molecular representations and categorized them into nine metaphor groups (see Fig. 2): color, clouds, ensemble, hulls, sausages, contours, texture, waves, and noise. We structured this section by grouping all methods according to their metaphor. Only a few of the collected methods were empirically evaluated [10], [11], [12], [13], [14]. We add more information about the evaluations within the context of their respective paper. Empirical evaluations of UV in a broader sense have been studied in various settings [15], [16], [17]. For an overview of evaluation methods and study designs, we refer to the survey by Hullman et al. [18].

In our work, we considered direct modifications of the representations. However, alternative methods for representing uncertainty information augment the visualization. Bryden et al. [19] used arrow glyphs to illustrate the consistent motion of larger groups of atoms. In MoFlow, Dabdoub et al. [10] used time series data to render pathlines of atoms to convey molecular motion. They conducted a preliminary study that suggests that MoFlow visualization helps users understand molecular motions better than traditional ensemble visualizations.

Another branch of UV for biomolecules includes uncertainty information in non-structural visualizations such as the Ramachandran plot, a 2D visualization of protein backbone torsion angles [9]. For more examples, we refer to Weiskopf, who described additional examples of UV for bioinformatics [20].

Next, we outline methods that directly modify the molecular representation grouped according to their metaphor.

Color Color is a very flexible metaphor for encoding physicochemical properties. It is often used to encode positional uncertainty [9], [21], [22], but other types of uncertainty can also be displayed. Ashford et al. [23], for example, visualized the probability of an atom or residue belonging to a pocket. The software Aquaria [24] infers the 3D structure of a protein based on its similarity to another protein's sequence (homology modeling). Modified colors in the representation encode the similarity of the sequences. Black signals unconserved parts, gray insertions, and muted colors signal uncertain components. Heinrich et al. [11] evaluated the color mapping in Aquaria in a crowdsourced study. They found a significant correlation between the perceived quality of images and alignment quality. Rasheed et al. [25], [26] derived uncertainties for quantities of interest such as electrostatic potential. They proposed mapping this uncertainty with colors to the surface. Schatz et al. [27] visualized the probability of a ligand reaching a point on the surface using color mapping. Color mapping is common for encoding uncertainty, but uncertainty is often only secondary information. Therefore, choosing another visual channel for UV is sometimes necessary to facilitate combined views of uncertainty and other quantities of interest.

Clouds Cloud visualizations often present volume data such as electron density maps [28]. But volumes can also be derived from atomistic data, e.g., by calculating the probability of a voxel being inside an atom sphere [25], [26]. Often, direct volume rendering is used for presenting molecules as clouds [8], [29], [30]. Knoll et al. [31] displayed two stacked clouds, where the surface of the inner cloud corresponds to the covalent radii and the surface of the outer one to the vdW radii. Related to cloud visualization are voxel maps, which present a wireframe visualization of feasible motions [32].

Ensemble Another popular option for displaying positional uncertainty is ensemble visualization, where several molecular conformations are aligned and superimposed [33], [34], [35], [36], [37]. Heinrich et al. [38] used parallel coordinates to select a subset of the ensemble by brushing. Usually, ensemble visualization uses abstract molecular representations such as backbone or cartoon visualizations. In contrast, Punjani et al. [39] created a series of isosurfaces from cryogenic-electron microscopy data. The surfaces are assigned individual colors and displayed simultaneously. In the previously discussed papers, all conformations are drawn opaquely. However, two papers investigated transparency as an option for displaying ensembles. Rheingans et al. [8] superimposed semi-transparent conformers. Another method was proposed by Melvin et al. [40], who rendered an opaque representative geometry and superimposed semi-transparent conformers. The superimposed conformers create a shadow- or cloud-like impression depending on the accumulation of opacity.

Hulls Hull visualizations represent uncertainty by rendering semi-transparent or opaque hulls around the molecule. Some methods display one semi-transparent hull by extracting an isosurface from volumetric data [8], [9], [29]. Other methods use several threshold values to create layers of semi-transparent hulls [41], [42], [43]. Goodsell et al. [29] proposed using several stacked opaque hulls. The individual levels can be seen as contour lines when the visualization is clipped at the image plane. Another option is the combination of conformations to form opaque surfaces. Fisher et al. [44] visualized a consensus surface based on two maximally displaced protein surfaces. Schmidt-Ehrenberg et al. [30] visualized the conformational density of an ensemble of up to three representative conformations using volume rendering and isosurfaces.

Sausages One popular variant of opaque hull visualizations for tube representations is the 'sausage' or 'putty' plot, where the tube width indicates the uncertainty [45], [46], [47].

Contours Sterzik et al. [12], [13] evaluated contours with modified line attributes for encoding biomolecular uncertainty. They used the line attributes blur, dashing, grayscale, sketchiness, and width to distinguish several uncertainty values on molecular representations, with width leading to consistently good results. Additionally, width was confirmed to be well distinguishable and popular for drawing uncertainty data in a more general evaluation of the line variables dashing, width, grayscale, and waves [48].

Texture Another option for encoding uncertainty through illustrative visualization [49] is the application of illustrative textures such as stippling or hatching, where the density encodes the uncertainty. Sterzik et al. [14] changed the primitive density according to the uncertainty. Through a perceptual study, they developed approximately perceptually uniform density levels for the illustrative textures. Closely related is the method of Lawonn et al. [50] which uses line integral convolution for highlighting salient regions and could also be used to highlight other properties such as uncertainty.

Waves Distortions of molecular representations can encode uncertainty. Schulz et al. [51] added sine waves to ribbon and tube visualizations to encode uncertainty. In their visualizations, they used the wave amplitude to encode the uncertainty.

Noise Geometrical distortions do not have to be regular like these proposed sine waves. Grigoryan and Rheingans [52] encoded uncertainty by randomly displacing surface points. Although not originally applied to molecular representations, we see their approach as a viable alternative.

IV. Representative Uncertainty Visualizations

In this section, we present representative visualizations for the metaphor categories discussed in the previous section. Given that evaluating all the previously proposed methods is infeasible, we selected these particular visualizations to represent the range of methods within each metaphor category effectively. In the following, we specify the design choices for the representatives used in our studies. In Fig. 2, we illustrate our representatives using the NMR ensemble of the protein 1ADN [53], [54]. We applied the uncertainty metaphors to three popular representations that effectively cover the range of available methods: the tube representation as an abstract model, the SES as a surface representation, and a stick visualization for atomic detail. Not all uncertainty metaphors can be applied to all three representations, e.g., the sausage visualization is specifically designed for the tube representation. Fig. 1 shows which uncertainty metaphors can be applied to each representation.

In this work, the metaphors visualize positional uncertainty derived from biomolecular ensembles. For scalar-based techniques, we selected the ensemble member closest to the average structure and computed RMSF values for each atomic position. These scalar RMSF values were then mapped onto the molecule using various visual encodings. For each molecule, we identified the minimum and maximum RMSF values and linearly mapped them to the respective visual encoding range (e.g., smallest to largest tube radius). This approach ensures that the full dynamic range of each visual variable is utilized per structure. In contrast, the cloud and ensemble metaphors depict the full conformational variability directly, without reducing it to a scalar summary per atom.

Color There exists no single standard colormap for encoding uncertainty in biomolecular visualizations. We chose increased lightness of our base color (blue) for increased uncertainty since lightness is a popular indicator for uncertainty [15], [55], [56]. The minimum RMSF observed across the structure is mapped to the base color (blue), and the maximum RMSF to white, with intermediate values linearly interpolated. The hue of a monochrome colormap seems to have little influence on its perception [57], [58].

Clouds For creating the cloud representation, we computed a density volume map based on the ensemble data with VMD [59] and rendered the clouds with a linear transfer function in ChimeraX [60]. Regions with higher density of conformations appear more opaque, and thus more transparent regions have higher positional uncertainty.

Ensemble For the ensemble representative, we rendered our aligned NMR ensemble directly. We chose to render all conformers opaquely since this is the most common approach in the literature. Uncertainty is conveyed through the visual spread and overlap of the conformers.

Hulls We rendered semi-transparent gray surfaces around the SES and the tube representations. The distance between the hull and the molecular surface encodes uncertainty.

Sausages This metaphor is based on the tube representation. The radius of the tube increases linearly with uncertainty, leading to thicker tubes in more uncertain regions.

Contours We rendered black contours around the molecular surface. The thickness of these contours increases proportionally with uncertainty.

Texture We used stippling density as the uncertainty attribute for the SES and hatching density for the tube representation. In both cases, higher uncertainty corresponds to lower texture density. To ensure more perceptually uniform variation across the uncertainty range, we applied the reparameterizations by Sterzik et al. [14].

Waves The wave representation (sinusoidal surface displacements) is based on the method of Schulz et al. [51], where larger amplitudes encode higher uncertainty. Frequency and direction of displacement are held constant to isolate amplitude as the primary uncertainty cue.

Noise We apply 3D Perlin noise to distort the SES and tube representations. The scalar noise is sampled based on spatial position and scaled by the uncertainty at each point. This produces spatially coherent surface variations, with larger displacements in more uncertain regions.

V. Qualitative Analysis With Domain Experts

As a first step in our evaluation of the uncertainty metaphors, we consulted six domain experts. Among them, three (SB1, SB2, SB3) specialize in structural biology with 23, 16, and 10 years of experience in the field. The remaining three (MS1, MS2, MS3) are experts in molecular simulations with 30, 29, and 6 years of experience. We conducted semi-structured interviews to explore the advantages and disadvantages of the metaphors. During the interviews, we presented representative visualizations for each metaphor one at a time. While the expert viewed the visualization, we asked whether they currently use or would consider using such visualizations in their work. We also inquired about their perspectives on the advantages and disadvantages of the visualizations. Based on their responses, we asked follow-up questions to explore specific points in greater depth. After discussing all metaphors, we concluded by asking for their opinions on key areas for future research.

A. Metaphors

Color All experts reported using color to display uncertainty information, but noted that the choice of colormap is nontrivial. The default red-white-blue (RWB) colormap found in tools like VMD [59] and ChimeraX [60] was uniformly rejected, as it is strongly associated with electrostatic potential and thus risks misinterpretation. SB1 reported using a green-orange colormap, and MS2 used a green-yellow-red scheme; however, both may present accessibility issues for color vision deficiencies. To address this, we chose a monochromatic colormap, which is common in UV across other domains [15], [56], [61]. SB2 appreciated this channel, noting that color is "flexible, can be applied to all representations, and is easy to grasp."

Clouds Feedback on the clouds was more ambiguous than for color. Three experts (SB1, SB2, SB3) stated they would not use cloud representations in publications, citing limited perceptual clarity in 2D and lack of familiarity within the community—e.g., SB2 noted that "the metaphor is not widely known", and SB1 commented that it can "look like a stain" on paper. MS1 also stressed that "the community wants to see concrete conformers."

However, in dynamic or interactive contexts (e.g., talks), most experts were more favorable. They noted that animation or explanation helps communicate how clouds result from overlapping conformations. MS3 was particularly enthusiastic, stating that "clouds are well suited for communicating that molecules are not as rigid as machines" and can be "easily interpreted as probability densities." MS1 added that cloud representations are important in quantum-chemical contexts or when ensembles are too large to visualize directly.

MS1 and MS2 noted they would combine clouds and color to highlight individual conformers or emphasize specific regions of interest within the cloud. Other experts emphasized simplicity: SB2 preferred minimal representations, stating that "the models

themselves are already very complex and it is necessary to focus on the important things." MS2 would only consider using clouds for space-filling representations and did not find them useful for abstract visualizations.

Ensemble Ensemble representations are the standard for NMR data, and all experts found them valuable for gaining a spatial understanding of conformational variability. However, their visual complexity received mixed responses. Although some appreciated the directness of the representation, SB3 described the visual result as an "absolute catastrophe" that "can't be shown to other people for demonstration purposes". MS2 also expressed reservations, noting they would only use ensembles for datasets with few conformers. SB2 proposed using color to highlight regions of structural equivalence across conformers to improve interpretability.

The experts also disagreed on the usefulness of ensemble visualizations for stick representations. Since a single stick model already introduces considerable visual clutter, ensemble overlays exacerbate this effect. SB2 and MS2 preferred to avoid such combinations altogether, whereas SB1 and SB3 found them helpful in specific cases, for example, to illustrate the flexibility of certain molecules.

Hulls Opinions on the hull metaphor were divided. The advantages of this representation include "reduced and therefore unobtrusive design" (MS1) and "capability to represent isotropic uncertainty" (SB2). MS1 also noted that hulls can be effective in "directing attention". In contrast, several aspects of the design received criticism. One concern was visual ambiguity: SB3 described the gray hull color as "confusing when placed on top the molecule's own color", and SB1 remarked that it provided "only additional color information when seen on the SES". MS3 mentioned the risk of misinterpretation, suggesting that hulls could be mistaken for a "water sheath around the molecule".

Expert feedback also pointed to questions of directionality and interaction possibilities. MS2 expressed a preference for inverting the encoding—wanting wider hulls in regions of lower uncertainty. SB2 proposed making the hull size interactive, suggesting a slider that could morph the hulls from a minimal indication of uncertainty to representing the full extent of the ensemble's conformational spread.

Sausages The sausage metaphor was generally well received, with four experts (SB1, SB3, MS1, MS3) appreciating its familiarity and established use in UV. MS1 specifically noted its "reduced design" as a strength. However, some limitations were mentioned. SB2 pointed out that the metaphor only applies to the tube representation, reducing its generalizability. MS2 highlighted a potential source of confusion: "in chemistry, wide and thin stands for back and front, respectively", which may interfere with intuitive interpretation. As with hulls, MS2 also expressed a preference for inverting the encoding—using wider elements to represent greater certainty instead of uncertainty.

Contours The contour metaphor was often compared to hulls. Two experts (MS1, MS2) preferred it over hulls, with MS1 highlighting its "reduced design" and MS3 noting that, unlike hulls, contours "can't be confused for water sheaths". However, MS3 also remarked that the metaphor "still does not imply uncertainty", pointing to a limitation in its expressiveness. SB2 considered contours sufficient for static images but inadequate for interactive 3D applications.

Texture The majority of experts (SB1, SB2, SB3, MS1, and MS3) found texture unconvincing as a representation of uncertainty. Several reasons were given: SB2 considered the textures distracting and unintuitive, while MS1 felt that hatching was not sufficiently prominent, and stippling involved too many point primitives. They reasoned that uncertainty, often a secondary data aspect, benefits from more subtle visual metaphors. Some experts raised concerns about unintended associations: SB1 noted that stippling could be misinterpreted as indicating importance rather than uncertainty, potentially guiding user attention in unintended ways. Similarly, hatching was seen as potentially confusable with structural details rather than uncertainty cues by SB1 and SB3.

In contrast, MS3 appreciated the aesthetic qualities of the texture visualizations but questioned their suitability for uncertainty communication, stating they "would be suited to visualize molecular properties, but do not look uncertain". MS2 was the only expert to endorse the texture metaphor positively for uncertainty, interpreting the white areas as possibilities and the blue areas as certain. However, they acknowledged that the actual spatial range accessible to the molecule was not directly depicted. MS2 favored applying the texture visualization specifically for surface (SES) and tube representations.

Waves The waves metaphor was broadly rejected by the experts, primarily due to concerns about misinterpretation. SB1, SB2, SB3, and MS2 noted that the sine wave pattern could be mistaken for actual molecular geometry rather than an uncertainty cue. Additionally, several experts (SB1, SB2, SB3) saw no perceptual or interpretive advantage over established representations such as sausages.

Noise Expert opinions on the noise metaphor were mixed. A recurring concern was that the visual effect resembled rendering or simulation artifacts rather than intentional uncertainty cues. SB1, SB3, and MS3 associated the noise with poor visual quality and found it aesthetically unconvincing. Some experts also found the noise metaphor only appropriate for certain representations. SB2 found it suitable only for the SES surface but not for tube representations, where it appeared like "bad quality". In contrast, MS2 had the opposite view, favoring the use of noise on tubes but not on SES surfaces. They argued that "on the SES, the noise does not cover the actual space that is accessible to the molecule. Therefore, the uncertain space might be underestimated." MS1 was more positive and reported that although the metaphor initially required some time to get used to, it ultimately conveyed uncertainty effectively.

B. Future Improvements

A commonly raised issue was the limited accessibility of UV (SB2, MS1, MS3). If at all, many of the metaphors are only accessible through specialized software. As a result, they are inconvenient or even inaccessible to most experts. Incorporating them as plugins into popular molecular visualization software suites could significantly broaden their usage. Additionally, MS3 advocated for such plugins to be capable of rendering only

specified interesting sites of the protein with the uncertainty metaphor. SB1 identified unresolved research problems regarding intrinsically disordered proteins (IDPs). In current cartoon representations, intrinsically disordered regions are depicted as lines, lacking an intuitive encoding for these highly uncertain areas. Additionally, in ensemble visualizations, the conformations do not have a uniform occupation density. Future work should focus on ensemble visualizations that account for this and illustrate the population of each conformation accurately. SB1 also emphasized the need for a more effective metaphor for flexible binding pockets, currently highlighted solely through colors. A visual analytics tool was requested by MS1. They stated that properly aligning ensembles of proteins is challenging, and thus, a tool that guides this process or suggests several versions of alignment would help. Furthermore, SB3 would like to see more abstract visualizations for uncertain data, similar to the illustrations by Goodsell [62], [63], [64]. Such abstract visualizations would avoid implying false accuracy, which is crucial for uncertainty representation.

C. Summary of Expert Feedback

To summarize, based on the conducted expert interviews, there is no superior uncertainty metaphor for proteins. Instead, preferences depended on both use case and representation type. Ensembles and clouds were valued primarily for exploring conformational space. While ensembles are the more established option, clouds were often seen as suitable for presentations. When uncertainty is treated as secondary information, reduced metaphors are better suited. Color coding was seen as familiar and easy to interpret, but few additional benefits were raised. Sausages were generally received well, mainly because they are already established. However, few specific advantages were named. Opinions on the newer metaphors—textures, noise, and waves—were more divided. These were often associated with unintended meanings: textures with structural or importance cues, noise with rendering artifacts, and waves with actual geometry. However, some experts pointed to specific use cases or reported that they came to appreciate these metaphors after some initial hesitation. There was also disagreement on the proper directionality for some metaphors. One expert suggested reversing the mapping in sausage and hull visualizations: using wider geometry for certain areas and thinner geometry for uncertain ones.

We did not observe major differences between structural biologists and molecular simulation experts. Structural biologists showed somewhat more aversion to the cloud metaphor, though this may reflect individual preferences, rather than a general trend within the discipline.

VI. Perceptual Evaluation

To ensure the objectivity of our results, we conducted three perceptual studies with a total of 1,756 participants. Fig. 3 shows an overview of the conducted perceptual studies. In the first and second studies, we addressed the most common disagreements between the experts: intuitiveness and directionality. For testing intuitiveness, we presented the visualizations to unbiased participants and asked them to describe the images. We tried

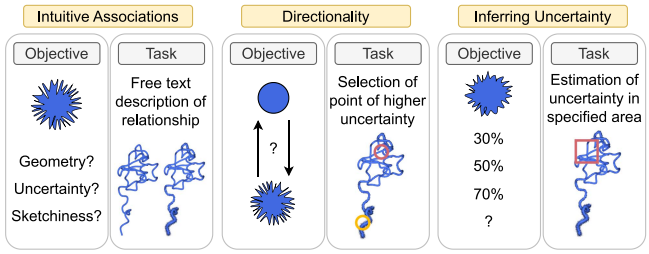


Fig. 3. Overview of the perceptual studies. Each study addresses a specific research question, illustrated by the accompanying figures. The first study evaluates whether the representative metaphors are intuitively associated with uncertainty. The second investigates whether their directionality is interpreted as intended. The third examines how accurately participants can infer uncertainty values from the metaphors. Illustrations of example tasks performed by participants are shown alongside each study objective.

to determine whether the visualizations spontaneously bring uncertainty to the mind. In the second study, we determined whether the directionality of the metaphors is appropriate, e.g., whether larger tube radii should encode uncertainty or certainty. In the last study, we tested how well uncertainty values can be inferred from the metaphors.

We received ethical approval for our research project from the ethical commission of the University of Jena. All participants consented to take part in this research project. We recruited participants using the crowdsourcing platform Prolific. The participants received a compensation of approximately £9 (∼10€ / $11) per hour. We deliberately recruited participants from the general public for our perceptual studies instead of relying on domain experts. Experts often exhibit biases towards familiar visualizations due to perceived cognitive ease [65]. As observed in our expert interviews, certain metaphors were more familiar to experts than others. By recruiting participants from the general public, we mitigated potential biases and created a more neutral testing environment. This approach ensures a fair evaluation, as experts might unknowingly favor suboptimal but familiar visual representations.

The participants were required to have normal or corrected-to-normal vision, no color vision deficiencies, and fluency in English. The participants were prompted not to participate on mobile devices and to maximize the browser window. Each participant only participated in one of the three studies to prevent bias. All stimuli and study responses are available on OSF at https://doi.org/10.17605/OSF.IO/DCTGU.

A. Intuitive Associations

The intuitiveness of the uncertainty metaphors repeatedly came up in the expert interviews. Often, the attributes that were described as intuitive were attributes that were already established in the molecular biology community. Newer visualizations were sometimes described as intuitive; however, for these newer metaphors, the expert opinions were divided.

Setup: We conducted this crowdsourced study with 540 participants to gain a more unbiased opinion on whether a metaphor spontaneously evokes a sense of uncertainty. The participants were shown two visualizations side-by-side. The visualization on the left side was a standard visualization of a protein; the visualization on the right encoded the uncertainty with one of

TABLE I
MENTIONS OF UNCERTAINTY OUT OF 540 RESPONSES

	Color	Clouds	Ensemble	Hulls	Sausages	Contours	Texture	Waves	Noise
Tube	1	2	0	2	1	0	1	1	3
SES	0	1	0	1	0	1	2	0	1
Sticks	0	0	0	0	0	0	0	0	0

the metaphors. The participants were prompted with: "On the left side, you see a visualization of a protein. On the right side, you see a different visualization of a protein. Please describe the relationship between the new visual elements and the protein." They were required to write at least 200 characters to submit their answers. We discussed the formulation of this task in-depth before conducting the study. The formulation was very open so as not to bias the participants. By providing a side-by-side view, we encouraged the participants to focus on the uncertainty metaphors, not the protein representation.

For variety, we used two NMR ensembles and one MD simulation. For each of these proteins, we generated 18 images. One image was created for each possible combination of the metaphors and representations. For applicable metaphors per representation, please refer to Fig. 1. We recruited ten participants per image; each participant saw one image.

Two of the authors independently reviewed the answers. We classified them into *describing uncertainty* or *not describing uncertainty*. If the two classifications differed, we discussed the individual answers and reached a consensus.

Results: For metaphors that clearly symbolize uncertainty, we expected a substantial number of answers that mention uncertainty or a synonym in the image description. However, out of the 540 answers we received, only 17 included a description of uncertainty. Most of the answers including an uncertainty description, were for noise (4 out of 17). Three answers were given for clouds, hulls, and texture, while color, contours, sausages, and waves each received one answer. We did not find any uncertainty answers for the ensemble metaphor. For a tabular breakdown of the responses by uncertainty metaphor and representation type, please refer to Table I. As our experts already hypothesized, the uncertainty metaphors sometimes got mistaken for additional structure like in this answer to the waves metaphor: "The second picture provides a more detailed visualization of [the] protein. It also shows amino acids and due to their size, we can imagine how [the] protein is located in space."

B. Directionality

For some of the metaphors, the intuitive directionality of the mapping is unclear. During the expert interviews, one expert repeatedly mentioned a preference for the opposite directionality, such as wider sausages representing more certain data. To investigate the suitability of the proposed mapping directionality, we conducted a second perceptual study.

Setup: The participants viewed UVs highlighting areas of minimal and maximal uncertainty and were asked to identify the area they believed had the higher uncertainty. Unlike in the initial study, participants here received a brief explanation of positional uncertainty in proteins. To confirm whether the participants used the uncertainty metaphor, they explained their strategy for identifying the more uncertain point in a separate question at the end of the study.

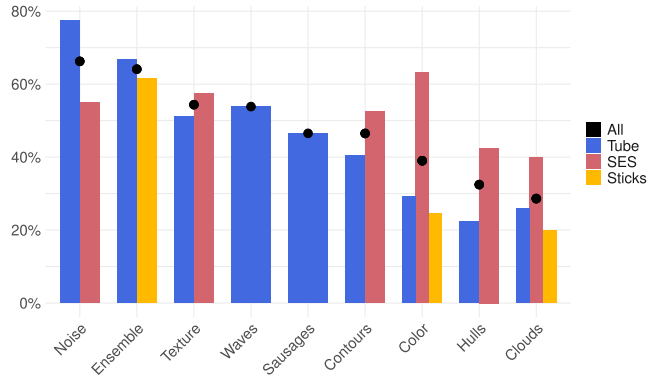


Fig. 4. Percentage of participants using the respective uncertainty attribute to identify the region of highest uncertainty when given a choice between the lowest and highest uncertainty areas. Black dots indicate the mean values over all representations.

We used nine proteins as stimuli, consisting of seven NMR ensembles and two MD simulations. The protein served as a within-participant variable, and to avoid potential biases, the uncertainty metaphor and protein representation were between-participant variables. Thus, each participant answered nine questions about the same uncertainty metaphor applied to the same molecular representation, with the order of proteins randomized. For each of the 18 groups—each possible combination of the uncertainty metaphors and molecular representations—we had two conditions for the number of participants: to recruit at least 20 participants, with at least 10 describing a strategy using the uncertainty metaphor. After three rounds of recruiting participants, we recruited sufficient participants for all groups, 793 in total.

Thus, in this study, we tested two properties of the metaphors: the intuitiveness when participants were aware they were assessing uncertainty and the dominant directionality of the mapping. If the metaphor clearly encodes uncertainty, we expected participants to use it to identify areas of higher uncertainty. In contrast to the first study, the participants were instructed to focus on uncertainty. Regarding directionality, we expected that most participants would correctly identify the area of greater uncertainty if the mapping direction was clear and intuitive. Two authors independently reviewed the explanations, resolving any disagreements through discussion until reaching a consensus.

Results: Fig. 4 shows the percentage of people who reported using the uncertainty metaphor. Noise and ensembles were used most frequently to judge uncertainty, each by over 60 % of participants. In contrast, hulls and clouds were the least used, with usage around 30 %. This showed that even when biased—i. e., prompted that this is an UV—some metaphors were not intuitive enough to be widely recognized as uncertainty encodings.

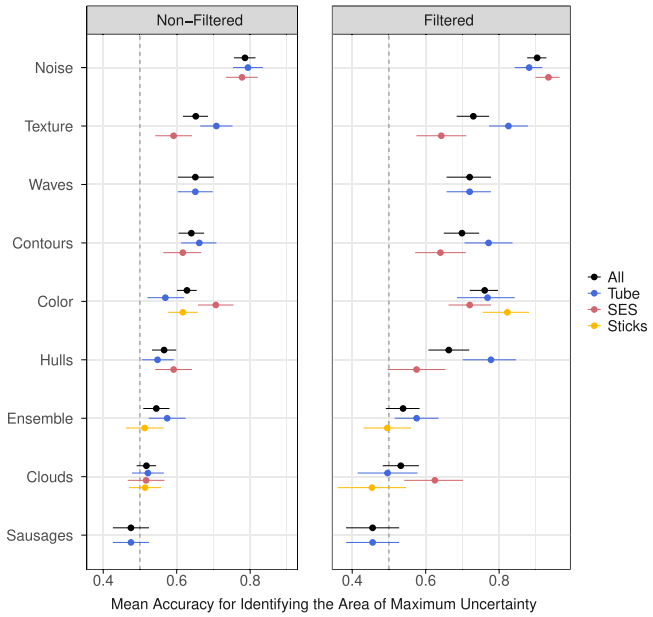


Fig. 5. Mean accuracy and 95% CI for identifying the area of maximum uncertainty when given a choice between an area of maximal and minimal uncertainty. The dashed line indicates an accuracy of 0.5. The left facet shows the non-filtered results, the right facet only includes participants that used the uncertainty attribute for their judgment.

The metaphors were used most often for the SES, followed by the tube, and least used for the stick representation. A notable exception was the noise metaphor, which was used about 20 percentage points more frequently for the tube representation compared to the SES. It might be easier to confuse noise for actual geometry on the SES as the tube representation is more cartoonish and abstract. For the SES, color was used most often to determine uncertainty: by more than 60% of participants. Surprisingly, for the tube and stick representations, color was only used by less than 30%.

Fig. 5 shows the mean accuracy for identifying the area of maximal uncertainty on a scale from 0 to 1. We report means and 95% confidence intervals (CI). If repeatedly computed from samples of the same distribution, 95,% of such intervals would contain the population mean [66]. CI are often preferred over null hypothesis significance testing as they reduce misinterpretations and dichotomous thinking [66], [67]. For metaphors where the directionality is clear and correctly assigned, we would expect an accuracy of approximately one. For incorrectly assigned directionality, we would expect an accuracy of approximately zero. However, we cannot conclude that for an accuracy of 0.5, the directionality was entirely unclear, as several confounding factors were involved. The most significant of these is likely that central areas tend to be more stable than those further from the center. We tried to include proteins where this is not immediately obvious. However, this only mitigates the problem slightly. Indeed, 115 participants reported looking for decentral parts of the protein as part of their strategy or deciding solely on that property.

Using only the answers filtered for using the uncertainty attribute increased the mean accuracy of all metaphors, apart from ensemble, clouds, and sausages, but the general order of accuracy between the metaphors stayed approximately the same. For noise, the directionality was the most obvious. In the filtered results, approximately 90% of the answers correctly identified the area of maximal uncertainty. This is consistent with the results from the intuitiveness study, where most of the uncertainty descriptions were given in response to the noise metaphor. The results for texture, waves, contours, color, and hulls were similar, with accuracies of approximately 0.7. For ensemble, clouds, and sausages, the accuracies were not considerably different from 0.5.

As previously mentioned, clouds and ensembles are more commonly used to gain an understanding of the conformational space than to estimate uncertainty values. However, for sausages, this indicates that the perception of the directionality is unclear, maybe even slightly skewed to being wrongly assigned if we factor in the confounding factor of centrality.

C. *Inferring Uncertainty Values*

After evaluating the intuitiveness and directionality of the uncertainty metaphors, we assessed how well uncertainty values can be inferred from the visualizations.

Setup: For this purpose, 423 participants estimated the uncertainties in highlighted areas of the proteins. For intuitive metaphors that clearly convey the level of uncertainty at specific positions on the molecule, we expect quick and highly accurate responses. To prevent learning effects, we divided the participants into 18 groups. Each group corresponds to a unique combination of metaphor and representation. Each participant had to estimate the uncertainty in 45 images (9 molecules $\times$ 5 areas). The participants were prompted to judge the uncertainty in the highlighted area from very certain to very uncertain on a 9-point Likert scale. The 9-point Likert scale covers a wide range of answer options while not being overwhelmingly detailed. To ensure that the Likert scale can be considered an interval scale, we used the common adjectives (“very”, “slightly”, etc.) for labeling the scale [68]. Each question also included an example visualization of a protein that was otherwise not part of the study. In the example image, the areas of highest and lowest uncertainty were highlighted. This was necessary for anchoring the Likert scale and for presenting the intended directionality of the uncertainty metaphor. An alternative to the example images would be providing a legend for the uncertainty mapping. However, since this is not feasible for the cloud and ensemble metaphor, the example images are more appropriate to ensure comparable conditions.

To analyze the ease of judgment, we timed the judgments. The participants were instructed to think about each question carefully while also trying to answer quickly. Following the 45 judgments, the participants assessed the visualizations across three categories using five-point Likert scales: visual appeal, level of confusion, and perceived task performance.

Stimuli generation: We created the stimuli from the same proteins as in the directionality study. For each protein, we generated five images with different areas highlighted. The areas had a size of 75×75 px for images of size 1024×1024 px. To ensure a balanced distribution of uncertainty values, we used a semi-automated process for identifying the areas.

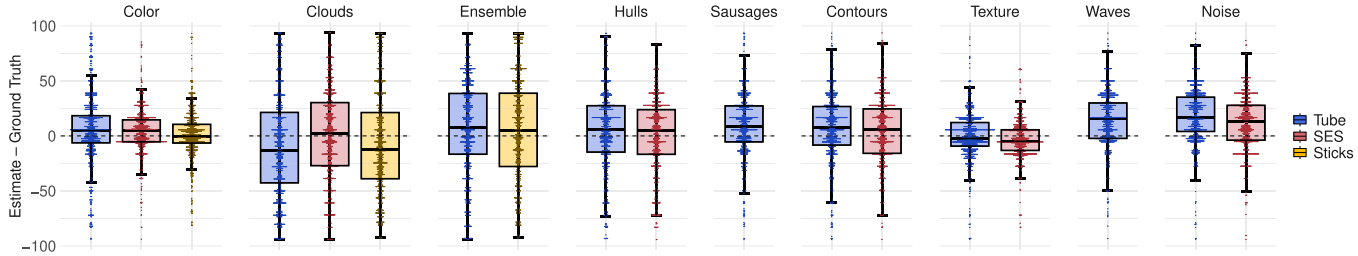


Fig. 6. Distributions of the participants' answers. The plots display the differences between the participant estimates and the ground truth. The boxplots summarize the distributions' key statistics: the black line within the box represents the median. The box indicates the interquartile range (IQR) with its length, spanning from the first (Q1) to the third (Q3) quartile. The whiskers extend to 1.5 IQR from each end of the box. The raw data values are overlaid as beeswarm plots.

For each protein image, we created a ground truth image showing the uncertainty values with a gray color map. We convolved each ground truth image with a filter of the area size and tracked three key metrics: mean uncertainty, uncertainty variance, and coverage. Coverage represents the ratio of protein-covered pixels to the total region size. To ensure adequate representation of protein in the highlighted area, we set coverage thresholds of 0.8 for SES and 0.2 for sticks and tubes. Next, we divided the range of uncertainty values into nine intervals and identified all area candidates meeting the required coverage for each interval. We then selected the region with minimal variance per interval to maintain relatively constant uncertainty values within the highlighted area. It was not always possible to find areas that satisfied every combination of the intervals and ground truth images. Thus, we settled on five intervals per ground truth image. Our selection criteria aimed for a uniform distribution of mean uncertainty values across all molecules and regions, with sufficient spacing between selected areas, high coverage, and low variance.

Analysis: Fig. 6 shows the differences between the participants' uncertainty estimates and the ground truths, aggregated per metaphor and representation. The differences were measured on a scale of uncertainty from 0 to 100%. In this context, the Likert scale responses were treated as the averages of nine intervals within this range. For instance, a Likert scale answer of 2 ("Certain") would correspond to approximately 16.67% uncertainty. If the ground truth of a stimulus was 50% uncertainty, and a participant judged it to be "Certain" the difference would thus be -33.33. Positive differences indicate an overestimation of the uncertainty, and negative differences indicate an underestimation. Given that participants were restricted to respond using a 9-point Likert scale, an individual response can only have a difference of zero if the ground truth aligns exactly with the midpoint of the chosen Likert scale interval. Our stimuli were not selected to coincide with these midpoints. Therefore, the differences would differ from zero even if a participant chose the option closest to the ground truth. However, over several measurements and participants, the average of the answers will approach zero if the participants can correctly identify the uncertainty level [68].

Results: For color and texture, most estimates cluster relatively close to the ground truth: the interquartile range (IQR) of differences remains below 25. In contrast, clouds and ensembles show the largest differences in both average magnitude and variability, with higher mean absolute differences and IQRs exceeding 55. This result aligns with expectations, as clouds and ensembles were developed to convey conformational variability rather than precise scalar uncertainty values.

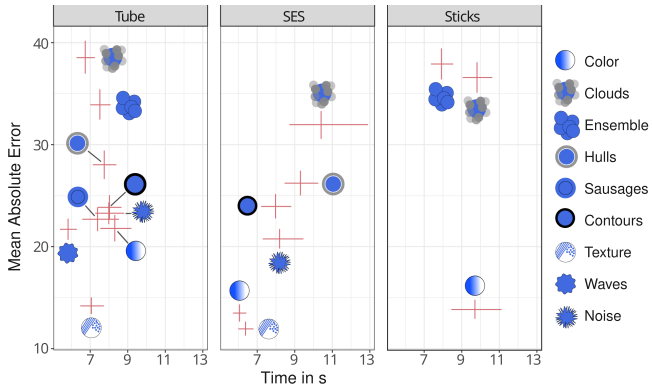


Fig. 7. MAE against the times required to answer. Mean and 95% CI are plotted in red and connected to metaphor icons.

Most metaphors tend to lead to overestimation of uncertainty (median differences > 2), except texture, where median differences are less than -2, and clouds, which cause underestimation on tubes and sticks (median differences < -12) but slight overestimation on the SES (median difference ~ 2). Color generally leads to overestimation, except on sticks where the bias is near zero. The estimation bias is highest for noise, with median differences greater than 13, especially on tube representation, where the median difference is 16.7 and more than 75% of responses, excluding outliers, overestimate uncertainty.

Fig. 7 shows the mean absolute error (MAE) plotted against the time it took the participants to answer the questions. Ideally, we would like the participants to be able to accurately and quickly infer uncertainty values from our visualizations.

For the tube representation, clouds result in the largest mean absolute errors (MAE 39) but relatively fast response times (7 s). Ensembles perform slightly better in terms of accuracy (MAE 34) but require more time (8 s). Sausages, noise, contours, and color achieve medium accuracy (MAE 23) with similar response times (8 s). Hulls (MAE 28) fall between this group and ensembles. Texture shows the best accuracy (MAE 14) and is also fast (7 s), while waves reach moderate accuracy (MAE 22) with the shortest response times (6 s). On the SES representation, texture and color produce the highest accuracy (MAE 13) and

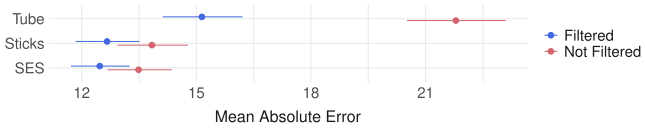


Fig. 8. Comparison of MAE for the color metaphor when excluding participants with seemingly random answer patterns and overall color MAE. We display means and 95% CI. Notably, for tube representations, a significant proportion of participants displayed such random patterns.

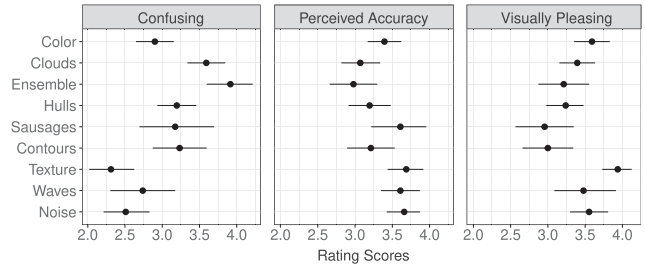


Fig. 9. Means and 95% CI of the ratings on confusion, perceived accuracy, and aesthetics.

the fastest response times (6 s). Noise, contours, and hulls show medium errors (MAE 24) and moderate response times (8 s). Clouds perform worse (MAE 32) and are slower (10 s). For the sticks, color performs best, with an MAE more than 20 units lower than that of clouds and ensembles. The response times are similar (around 9 s), with ensembles being slightly faster.

For all metaphors except color, the results are similar across all representations. In contrast, color shows notably worse performance on the tube representation, with MAE approximately eight percentage points higher than on SES or sticks. We investigated this discrepancy further by examining the results of individual participants. We observed that, for most metaphors, there were few instances of participants exhibiting seemingly random answer distributions, such as uniform responses regardless of uncertainty value or random switching between minimal and maximal uncertainty. In contrast, many participants in the clouds and ensemble groups submitted seemingly random responses. The overall accuracy for these metaphors was notably low—at least nine units higher in MAE than all others—suggesting that the associated tasks were particularly difficult for participants to judge. In contrast, the other metaphors yielded higher overall accuracy. For these, we believe that seemingly random responses are more likely due to a small number of participants not following the instructions. Such instances were relatively evenly distributed across representations. However, for the color metaphor, the tube representation appeared to have a notably higher frequency of random submissions. We filtered the participants assigned to the color metaphor based on the correlation coefficients of their answers. Six of the 24 participants assigned to the tube representation had negative correlation coefficients. For comparison, only 1 in 23 participants had negative correlation coefficients for sticks and SES each.

When recomputing the MAE without those randomly answering participants, the results for the three representations are much more similar. All data points—including those appearing random—are depicted in Figs. 6 and 7. A comparison of MAE computed with and without these seemingly random responses is depicted in Fig. 8. When excluding the random answers, the MAE of the tube-color group is still higher than for texture but smaller than for the cluster of contours, sausages, waves, and noise. It is unclear why about 25% of the participants in the tube-color group answered this randomly. In future work, a closer evaluation of color perception differences on the representations would be interesting.

Participant Ratings: Fig. 9 shows the ratings of the metaphors by the participants. The participants seemed to find the texture, noise, and waves metaphors the least confusing and the clouds and ensemble representations the most confusing. Even so, they rated their accuracy relatively similar for all metaphors. Texture, noise, and waves are rated slightly better than ensemble and hulls, but the differences are small. Texture was perceived to be the most visually pleasing. The differences between the other metaphors were marginal.

D. Summary of the Perceptual Evaluation

In our intuitiveness study, only few responses described uncertainty, indicating that none of the visualization metaphors strongly evoke it. As the study was intentionally open so as not to bias the participant, further research is needed to detect weaker associations. Our next study gave additional insights into intuitiveness. In this setup, we explicitly informed the participants that the visualizations encoded uncertainty. If a metaphor naturally evoked associations with uncertainty, we would expect most participants to use the metaphor to solve the given task. The results reinforced our earlier conclusion: none of the visualization metaphors are strongly associated with uncertainty. Across all metaphors, fewer than 70% of participants employed the metaphor to complete the task, and for most metaphors, usage was below 50%.

Regarding directionality, our results suggest that most metaphors use an appropriate mapping direction. However, apart from the noise metaphor, the other variables had less than 80% accuracy in correctly identifying the directionality. Therefore, proper labeling and/or training of the participants remains crucial. The sausage metaphor is an exception. With less than 50% accuracy, the directionality of this metaphor seems unintuitive.

In the last study, participants were less accurate in inferring uncertainty values from the conformational space visualizations. This was expected since the other representations visualize the scalar uncertainty value directly. In conformational space visualizations, viewers must interpret and abstract the information on their own.

VII. Discussion

In the following, we discuss our evaluations in more detail.

A. Conformational Space Visualizations

According to the interviewed experts, ensembles are of utmost importance for a spatial understanding of the data. However, ensemble visualizations tend to get crowded when the number of molecular conformations becomes large. Clouds are more uncommon. They also give an impression of the conformational

space and are suitable for large ensembles. In fact, they even benefit from more data. However, they are not established and need thorough explanations to be understood. Clouds and ensembles both consistently lead to poor results in our perceptual experiments. However, since the tasks were more targeted towards estimating uncertainty values, this outcome was somewhat expected. Surprisingly, these metaphors did not even seem to help identify the area of higher uncertainty from a choice of the areas with minimum and maximum uncertainty. Additionally, neither was intuitively perceived as visualizing uncertainty or possible conformational spaces. Thus, the metaphors do not seem to be self-explanatory. We suspect this might be due to them getting crowded and overwhelming. Some work has already been done to improve the clarity of ensembles. Brushing and linking in an interactive tool [38] and selective filtering of conformers combined with transparency [40] are promising approaches that need further evaluation regarding their perception. In general, a more reduced or stylized visualization of the whole conformational space might help while still providing a spatial understanding.

B. Representation of Summary Statistics

Hulls can be used to visualize conformational spaces by visualizing the probability density by rendering multiple isosurfaces. However, they are commonly used to indicate uncertainty levels with only one isosurface. We evaluated the latter option. The hulls received mixed feedback from our experts. They were liked for the possibility of displaying anisotropic uncertainty information and their reduced design, but the experts stated the hulls might be confused for additional elements. For inferring values, the hulls had slightly worse results than similarly reduced metaphors such as sausages and contours. One of our experts wished for an adjustable version of hulls where the extent of the hulls could be modified with a slider, from a small hull serving as an indication of uncertainty to the maximal hull of all conformations.

The sausages were valued by most of our experts as being intuitive for visualizing uncertainty information. Contrarily, our perceptual evaluation showed that the directionality of the mapping—large radii for uncertain data—is likely not perceived as intended. The sausages metaphor was the only metaphor where the directionality seems counter-intuitive, with less than 50% accuracy on average. Contours are an alternative to sausages that the experts found similar. In the perceptual evaluations, contours performed similarly to sausages, but their directionality seemed more intuitive to the participants.

For noise, the directionality was clearest; approximately 90% of the answers were correct. Additionally, noise seemed to be the most intuitive as it was most frequently used to determine the point of maximal uncertainty across all metaphors in the directionality study. Thus, noise seemed to be most easily recognizable as an uncertainty encoding. The absolute errors in inferring uncertainty values for noise were relatively small. However, the estimations exhibited the highest bias towards overestimating uncertainty among all uncertainty metaphors. This might be due to noise most prominently encoding uncertainty. Contrary to the expert opinions, most participants found the noise visualizations visually pleasing. Although closely related to the noise metaphor, waves appeared less intuitive as they were used less frequently to determine the maximal uncertainty. Furthermore, the directionality was less clear than that of noise. For inferring uncertainty values, the waves metaphor had similar accuracy to noise but the fastest response times out of all uncertainty metaphors. The waves visualization was disliked by experts, who stated it could easily be confused with actual geometry. Although our study did not explicitly evaluate whether metaphors that manipulate geometry, such as waves, are more likely to be misinterpreted as molecular structure, we observed examples of such misinterpretation in the free-text responses of our directionality study. This suggests that metaphors with geometrically salient positional variation may be more prone to misinterpretation. While such confusion might be mitigated through viewer training or explanatory legends, this type of metaphor demands careful design consideration, particularly in scientific communication contexts.

Color was the most widely used attribute for encoding scalar uncertainty values. The primary disadvantage of color mapping is that the color channel cannot be used to represent other attributes. As uncertainty is often secondary information, this can be problematic. Apart from this, color was liked by the experts for being a flexible metaphor that can be used for any representation and is easily available. The directionality of the mapping—increased lightness for increased uncertainty—seemed to be suitable. Color also proved effective for inferring uncertainty values. However, we noticed that for judgments of uncertainty on the tube representation, about 25% of the participants seemed to answer randomly, considerably more than for the other representations. This effect needs further examination in future work.

Only one of the interviewed experts found texture to be a convincing metaphor for uncertainty, but in the perceptual studies, texture performed well. The percentage of people who used texture to determine the uncertainty in the directionality study was higher than for most other metaphors. The directionality of the mapping (low density for uncertain values) also worked well. Texture had the lowest absolute errors for inferring uncertainty values and was judged to be the least confusing and the most visually pleasing out of all metaphors.

C. Limitations

Our evaluations provided insights into the appropriateness of various uncertainty metaphors for biomolecular structure visualization. However, there are some limitations. We chose representatives that cover the range of proposed uncertainty metaphors. Despite this, variations within individual categories cannot be fully addressed in a single set of studies. While our results offered indications regarding the suitability of the categories, follow-up research should explore multiple variants within each category (e.g., different types of uncertainty hulls). Furthermore, our studies were conducted with 2D images, and the results might differ in interactive or stereoscopic 3D environments. This is particularly relevant for the hulls and clouds visualizations, which experts believed would perform better in 3D. While static 2D images represent a common and practical use case, future

work should also explore how these visualizations are perceived in dynamic and immersive 3D settings, for example, by investigating multiple viewing angles and interactive environments. In this evaluation, uncertainties were always compared within one molecule. Thus, the whole range of the uncertainty metaphor could be utilized for each protein. When comparing results between several proteins, the uncertainty attributes might need to encode larger variances in uncertainty, which could change the results.

D. Future Work

In our perceptual studies, we evaluated the appropriateness of the metaphors in the first two studies. The third study investigated the effectiveness by assessing how accurately uncertainty values can be inferred from the metaphors. To further assess the effectiveness of the metaphors, future work should examine how uncertainty metaphors might influence domain-specific interpretations or conclusions. Our current evaluation focused on expert preferences and the perception of metaphors. Understanding how such visualizations affect scientific reasoning in applied contexts would require domain-specific evaluations involving concrete scientific tasks or decision-making scenarios.

While our studies focused on structural molecule data, our results might apply to more general use cases. Many of the metaphors are relevant to other types of data, and the questions posed in studies 2 and 3 are easily transferable. In future work, it will be important to assess the generalizability of our results using a broader range of datasets. Similarly, insights from other application areas of UV could apply to biomolecular structures. For example, sonification and animation techniques such as blinking or oscillation [69] remain unexplored in this context. Another form of animation—Hypothetical Outcome Plots—animate finite draws from a probability distribution [70]. This could be a promising technique for biomolecular structures. Furthermore, techniques for visualizing data distributions, such as Gaussian mixture models [71], could be tested for representing conformational spaces. Exploring these approaches may provide new strategies for UV in biomolecular applications.

Additionally, our experts mentioned several areas of molecular UV that require future work. They mentioned intuitive representations for intrinsically disordered regions in proteins, accounting for the non-uniform distribution of conformations in molecular ensembles, visual analytics guidance for aligning protein ensembles, effective metaphors for flexible binding pockets, and more abstract, illustrative visualizations for proteins. However, the most critical area for future work is increasing the accessibility of UV. While all the metaphors evaluated in our studies have been proposed for UV before, only a few are readily accessible to practitioners. This severely hinders their widespread usage. Providing plugins for the commonly used molecular visualization tools could drastically increase the usage of UV for biomolecular structures.

E. Key Insights

Our analysis of visualization methods, based on expert interviews and perceptual studies, revealed several key insights. First, no single visualization method can address all challenges equally well, as each has distinct strengths and limitations. As a result, there is no universally recommended approach for UV. Instead, the choice of metaphor should be guided by the specific requirements of the task. For example, conformational space visualizations, such as ensembles and cloud visualizations, are particularly effective for providing a broad overview of the entire conformational space. On the other hand, metaphors that represent summary statistics of uncertainty distributions are better suited for tasks where uncertainty is a secondary consideration, offering a more reduced representation.

The study also highlighted the influence of expert biases on the choice of visualization metaphor. Familiar representations, such as ensembles, sausages, or color, were generally rated more positively by experts compared to less familiar approaches. However, the perceptual study results showed that some less conventional metaphors, such as noise and texture, performed well, indicating potential value in exploring these alternatives. This suggests that established methods may not always be the most effective and should be assessed critically rather than automatically favored.

One notable discrepancy between common practice and our study results concerns the sausage metaphor. While this metaphor is commonly used to encode uncertainty with larger radii, our perceptual study found that more than 50% of participants interpreted larger radii as representing higher certainty. This suggests that the typical mapping direction of the sausage metaphor might be inappropriate. When using this metaphor to represent uncertainty, it is crucial to provide clear guidance to ensure users understand the intended meaning.

VIII. Summary and Conclusion

We collected and categorized visualization methods for positional uncertainty in proteins. We selected nine representative visualization metaphors and evaluated them through expert interviews and three crowdsourced studies. The metaphors have different strengths, weaknesses, and application areas. Ensembles and clouds provide spatial overviews. Experts rely on this to understand the protein's conformational space. However, our perceptual studies showed that these visualizations tend to be confusing and unintuitive. Some metaphors, like contours, hulls, and sausages, offer a very reduced design and thus add uncertainty information to visualizations in a particularly unobtrusive way. Among these very reduced metaphors, we recommend using contours. They appeared more intuitive than sausages and provided better accuracy for inferring uncertainty values compared to hulls. Metaphors that are neither very reduced nor display the whole conformational space are color, texture, waves, and noise. Noise seemed most easily recognized as an uncertainty attribute but led to overestimated uncertainty values. We recommend using the metaphor noise to create easy-to-understand visualizations where the accuracy of estimating specific uncertainty values is not crucial. Texture was best for inferring uncertainty values and only slightly less intuitive than noise. Participants frequently recognized texture as encoding uncertainty, and the directionality of the mapping seems intuitive. Thus, texture emerged as the most well-rounded metaphor in our studies.

Supplemental Material Pointer

We made the stimuli and participant responses of all three perceptual studies available on OSF at https://doi.org/10.17605/OSF.IO/DCTGU under a CC BY 4.0 license.

Acknowledgment

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References

[1] B. Kozlíková et al., "Visualization of biomolecular structures: State of the Art revisited," *Comput. Graph. Forum*, vol. 36, no. 8, pp. 178–204, 2017.
[2] G.-P. Bonneau et al., "Overview and state-of-the-art of uncertainty visualization," in *Scientific Visualization*. Berlin, Germany: Springer, London, 2014, pp. 3–27.
[3] C. Johnson, "Top scientific visualization research problems," *IEEE Comput. Graph. Appl.*, vol. 24, no. 4, pp. 13–17, Jul./Aug. 2004.
[4] A. Sterzik, C. Gillmann, M. Krone, and K. Lawonn, "Uncertainty-aware visualization of biomolecular structures," in *Computer Graphics Forum*. New York, NY, USA: Wiley, 2025, Art. no. e70155.
[5] F. U. Hartl, "Protein misfolding diseases," *Annu. Rev. Biochem.*, vol. 86, pp. 21–26, 2017.
[6] K. Henzler-Wildman and D. Kern, "Dynamic personalities of proteins," *Nature*, vol. 450, no. 7172, pp. 964–972, 2007.
[7] H. J. Dyson and P. E. Wright, "Intrinsically unstructured proteins and their functions," *Nature Rev. Mol. Cell Biol.*, vol. 6, no. 3, pp. 197–208, 2005.
[8] P. Rheingans and S. Joshi, "Visualization of molecules with positional uncertainty," in *Proc. VisSym99: Joint Eurographics IEEE TCVG Symp. Visual.*, 1999, pp. 299–306.
[9] R. G. Maack, M. L. Raymer, T. Wischgoll, H. Hagen, and C. Gillmann, "A framework for uncertainty-aware visual analytics of proteins," *Comput. Graph.*, vol. 98, pp. 293–305, 2021.
[10] S. M. Dabdoub, R. Rumpf, A. D. Shindhelm, and W. C. Ray, "MoFlow: Visualizing conformational changes in molecules as molecular flow improves understanding," in *BMC Proc.*, vol. 9, 2015, Art. no. S5.
[11] J. Heinrich, S. Kaur, and S. O'Donoghue, "Evaluating the effectiveness of color to convey alignment quality in macromolecular structures," in *Proc. 2015 Big Data Vis. Analytics*, 2015, pp. 1–8.
[12] A. Sterzik, N. Lichtenberg, M. Krone, D. W. Cunningham, and K. Lawonn, "Perceptual evaluation of common line variables for displaying uncertainty on molecular surfaces," in *Proc. Eurographics Workshop Vis. Comput. Biol. Med.*, 2022, pp. 41–51.
[13] A. Sterzik, N. Lichtenberg, M. Krone, D. Baum, D. W. Cunningham, and K. Lawonn, "Enhancing molecular visualization: Perceptual evaluation of line variables with application to uncertainty visualization," *Comput. Graph.*, vol. 114, pp. 401–413, 2023.
[14] A. Sterzik, M. Meuschke, D. W. Cunningham, and K. Lawonn, "Perceptually uniform construction of illustrative textures," *IEEE Trans. Vis. Comput. Graph.*, vol. 30, no. 1, pp. 1052–1062, Jan. 2024.
[15] J. Sanyal, S. Zhang, G. Bhattacharya, P. Amburn, and R. J. Moorhead, "A user study to compare four uncertainty visualization methods for 1D and 2D datasets," *IEEE Trans. Vis. Comput. Graph.*, vol. 15, no. 6, pp. 1209–1218, Nov./Dec. 2009.
[16] N. Boukhelifa, A. Bezerianos, T. Isenberg, and J. D. Fekete, "Evaluating sketchiness as a visual variable for the depiction of qualitative uncertainty," *IEEE Trans. Vis. Comput. Graph.*, vol. 18, no. 12, pp. 2769–2778, Dec. 2012.
[17] S. Tak, A. Toet, and J. van Erp, "The perception of visual uncertainty representation by non-experts," *IEEE Trans. Vis. Comput. Graph.*, vol. 20, no. 6, pp. 935–943, Jun. 2014.
[18] J. Hullman, X. Qiao, M. Correll, A. Kale, and M. Kay, "In pursuit of error: A survey of uncertainty visualization evaluation," *IEEE Trans. Vis. Comput. Graph.*, vol. 25, no. 1, pp. 903–913, Jan. 2019.
[19] A. Bryden, G. N. Phillips, and M. Gleicher, "Automated illustration of molecular flexibility," *IEEE Trans. Vis. Comput. Graph.*, vol. 18, no. 1, pp. 132–145, Jan. 2012.
[20] D. Weiskopf, "Uncertainty visualization: Concepts, methods, and applications in biological data visualization," *Front. Bioinf.*, vol. 2, 2022, Art. no. 793819.
[21] C.-D. Zachmann, S. M. Kast, and J. Brickmann, "Quantification and visualization of molecular surface flexibility," *J. Mol. Graph.*, vol. 13, no. 2, pp. 89–97, 1995.
[22] M. F. Thorpe, M. Lei, A. J. Rader, D. J. Jacobs, and L. A. Kuhn, "Protein flexibility and dynamics using constraint theory," *J. Mol. Graph. Modelling*, vol. 19, no. 1, pp. 60–69, 2001.
[23] P. Ashford et al., "Visualisation of variable binding pockets on protein surfaces by probabilistic analysis of related structure sets," *BMC Bioinf.*, vol. 13, 2012, Art. no. 39.
[24] S. I. O'Donoghue et al., "Aquaria: Simplifying discovery and insight from protein structures," *Nature Methods*, vol. 12, no. 2, pp. 98–99, 2015.
[25] M. Rasheed, N. Clement, A. Bhowmick, and C. Bajaj, "Statistical framework for uncertainty quantification in computational molecular modeling," in *Proc. 7th ACM Int. Conf. Bioinf., Comput. Biol., Health Inform.*, 2016, pp. 146–155.
[26] M. Rasheed, N. Clement, A. Bhowmick, and C. L. Bajaj, "Statistical framework for uncertainty quantification in computational molecular modeling," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 16, no. 4, pp. 1154–1167, Jul./Aug. 2019.
[27] K. Schatz et al., "Visual analysis of large-scale protein-ligand interaction data," *Comput. Graph. Forum*, vol. 40, no. 6, pp. 394–408, 2021.
[28] M. Hu, W. Chen, T. Zhang, and Q. Peng, "Direct volume rendering of volumetric protein data," in *Proc. Adv. Comput. Graph.*, ser. Lecture Notes in Computer Science, 2006, pp. 397–403.
[29] D. S. Goodsell, I. Mian, and A. J. Olson, "Rendering volumetric data in molecular systems," *J. Mol. Graph.*, vol. 7, no. 1, pp. 41–47, 1989.
[30] J. Schmidt-Ehrenberg, D. Baum, and H.-C. Hege, "Visualizing dynamic molecular conformations," in *Proc. IEEE Visual.*, 2002, pp. 235–242.
[31] A. Knoll et al., "Uncertainty classification and visualization of molecular interfaces," *Int. J. Uncertainty Quantification*, vol. 3, no. 2, pp. 157–169, 2013.
[32] J. Cortés, S. Barbe, M. Erard, and T. Siméon, "Encoding molecular motions in voxel maps," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 8, no. 2, pp. 557–563, Mar./Apr. 2011.
[33] L. A. Kelley and M. J. Sutcliffe, "OLDERADO: On-line database of ensemble representatives and domains," *Protein Sci.: A Pub. Protein Soc.*, vol. 6, no. 12, pp. 2628–2630, 1997.
[34] C. Best and H.-C. Hege, "Visualizing and identifying conformational ensembles in molecular dynamics trajectories," *Comput. Sci. Eng.*, vol. 4, no. 3, pp. 68–75, 2002.
[35] P. Lai, W. Kaplan, W. B. Church, and R. K. Wong, "Informative 3D visualization of multiple protein structures," in *Proc. 2nd Conf. Asia-Pacific Bioinf.*, 2004, pp. 201–208.
[36] F. S. Domingues, J. Rahnenführer, and T. Lengauer, "Conformational analysis of alternative protein structures," *Bioinformatics*, (Oxford, England), vol. 23, no. 23, pp. 3131–3138, 2007.
[37] A. J. M. Martin, I. Walsh, and S. C. E. Tosatto, "MOBI: A web server to define and visualize structural mobility in NMR protein ensembles," *Bioinformatics*, (Oxford, England), vol. 26, no. 22, pp. 2916–2917, 2010.
[38] J. Heinrich, M. Krone, S. I. O'Donoghue, and D. Weiskopf, "Visualising intrinsic disorder and conformational variation in protein ensembles," *Faraday Discuss.*, vol. 169, pp. 179–193, 2014.
[39] A. Punjani and D. J. Fleet, "3DFlex: Determining structure and motion of flexible proteins from cryo-EM," *Nature Methods*, vol. 20, no. 6, pp. 860–870, 2023.
[40] R. L. Melvin and F. R. Salsbury, "Visualizing ensembles in structural biology," *J. Mol. Graph. Modelling*, vol. 67, pp. 44–53, 2016.
[41] C. H. Lee and A. Varshney, "Representing thermal vibrations and uncertainty in molecular surfaces," in San Jose, CA, USA: *Electron. Imag.*, 2002, 2002, pp. 80–90.
[42] M. Krone, K. Bidmon, and T. Ertl, "Interactive visualization of molecular surface dynamics," *IEEE Trans. Vis. Comput. Graph.*, vol. 15, no. 6, pp. 1391–1398, Nov./Dec. 2009.
[43] T. Ertl, M. Krone, S. Kesselheim, K. Scharnowski, G. Reina, and C. Holm, "Visual analysis for space–time aggregation of biomolecular simulations," *Faraday Discuss.*, vol. 169, pp. 167–178, 2014.
[44] C. L. Fisher, J. A. Tainer, M. E. Pique, and E. D. Getzoff, "Visualization of molecular flexibility and its effects on electrostatic recognition," *J. Mol. Graph.*, vol. 8, no. 3, pp. 125–132, 145, 1990.
[45] R. Koradi, M. Billeter, and K. Wüthrich, "MOLMOL: A program for display and analysis of macromolecular structures," *J. Mol. Graph.*, vol. 14, no. 1, pp. 51–55, 1996.
[46] W. R. P. Scott and S. K. Straus, "Determining and visualizing flexibility in protein structures," *Proteins: Struct., Funct., Bioinf.*, vol. 83, no. 5, pp. 820–826, 2015.
[47] A. Jurčík, J. Parulek, J. Sochor, and B. Kozlikova, "Accelerated visualization of transparent molecular surfaces in molecular dynamics," in *Proc. 2016 IEEE Pacific Visual. Symp.*, 2016, pp. 112–119.

[48] A. Sterzik, N. Lichtenberg, J. Wilms, M. Krone, D. W. Cunningham, and K. Lawonn, "Perception of line attributes for visualization," *IEEE Trans. Vis. Comput. Graph.*, vol. 30, no. 1, pp. 1041–1051, Jan. 2024.

[49] K. Lawonn, I. Viola, B. Preim, and T. Isenberg, "A survey of surface-based illustrative rendering for visualization," *Comput. Graph. Forum*, vol. 37, no. 6, pp. 205–234, 2018.

[50] K. Lawonn, M. Krone, T. Ertl, and B. Preim, "Line integral convolution for real-time illustration of molecular surface shape and salient regions," *Comput. Graph. Forum*, vol. 33, no. 3, pp. 181–190, 2014.

[51] C. Schulz, K. Schatz, M. Krone, M. Braun, T. Ertl, and D. Weiskopf, "Uncertainty visualization for secondary structures of proteins," in *2018 IEEE Pacific Visual. Symp.*, 2018, pp. 96–105.

[52] G. Grigoryan and P. Rheingans, "Point-based probabilistic surfaces to show surface uncertainty," *IEEE Trans. Vis. Comput. Graph.*, vol. 10, no. 5, pp. 564–573, Sep./Oct. 2004.

[53] L. C. Myers, G. L. Verdine, and G. Wagner, "Solution structure of the DNA methyl phosphotriester repair domain of Escherichia coli ADA," *Biochemistry*, vol. 32, no. 51, pp. 14089–14094, 1993.

[54] L. Myers, G. Verdine, and G. Wagner, "Solution structure of the DNA methylphosphotriester repair domain of Escherichia coli ADA," 1994, doi: 10.2210/pdb1adn/pdb.

[55] B. Jiang, A. Brown, and F. Ormeling, "Some perceptual aspects of colouring uncertainty," in *Proc. Adv. GIS Res. II Int. Symp. Spatial Data Handling*, 1997, pp. 29–42.

[56] A. M. MacEachren, R. E. Roth, J. O'Brien, B. Li, D. Swingley, and M. Gahegan, "Visual semiotics & uncertainty visualization: An empirical study," *IEEE Trans. Vis. Comput. Graph.*, vol. 18, no. 12, pp. 2496–2505, Dec. 2012.

[57] S. Lewandowsky, D. J. Herrmann, J. T. Behrens, S.-C. Li, L. Pickle, and J. B. Jobe, "Perception of clusters in statistical maps," *Appl. Cogn. Psychol.*, vol. 7, no. 6, pp. 533–551, 1993.

[58] Y. Liu and J. Heer, "Somewhere over the rainbow: An empirical assessment of quantitative colormaps," in *Proc. 2018 CHI Conf. Hum. Factors Comput. Syst.*, New York, NY, USA, 2018, pp. 1–12.

[59] W. Humphrey, A. Dalke, and K. Schulten, "VMD–visual molecular dynamics," *J. Mol. Graph.*, vol. 14, pp. 33–38, 1996.

[60] E. F. Pettersen et al., "UCSF ChimeraX: Structure visualization for researchers, educators, and developers," *Protein Sci.: A Pub. Protein Soc.*, vol. 30, no. 1, pp. 70–82, 2021.

[61] M. Correll, D. Moritz, and J. Heer, "Value-suppressing uncertainty palettes," in *Proc. 2018 CHI Conf. Hum. Factors Comput. Syst.*, 2018, pp. 1–11.

[62] D. S. Goodsell and J. Jenkinson, "Molecular illustration in research and education: Past, present, and future," *J. Mol. Biol.*, vol. 430, no. 21, pp. 3969–3981, 2018.

[63] D. S. Goodsell, L. Autin, and A. J. Olson, "Illustrate: Software for biomolecular illustration," *Structure* (London, England : 1993), vol. 27, no. 11, pp. 1716–1720.e1, 2019.

[64] D. S. Goodsell, C. Zardecki, H. M. Berman, and S. K. Burley, "Insights from 20 years of the molecule of the month," *Biochem. Mol. Biol. Educ.*, vol. 48, no. 4, pp. 350–355, 2020.

[65] A. Dasgupta, "Experts' familiarity versus optimality of visualization design: How familiarity affects perceived and objective task performance," in *Cognitive Biases in Visualizations*. Berlin, Germany: Springer, 2018, pp. 75–86.

[66] M. J. Gardner and D. G. Altman, "Confidence intervals rather than p values: Estimation rather than hypothesis testing," *BMJ* (Clinical research ed.), vol. 292, no. 6522, pp. 746–750, 1986.

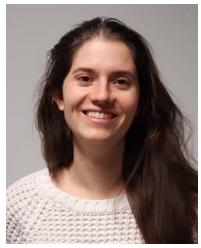
[67] P. Dragicevic, "Fair statistical communication in HCI," in *Modern Statistical Methods for HCI*, Berlin, Germany: Springer, 2016, pp. 291–330.

[68] D. Cunningham and C. Wallraven, *Experimental Design: From User Studies to Psychophysics, 1st* ed. Natick, MA, USA: A. K. Ltd Peters, 2011.

[69] A. T. Pang, C. M. Wittenbrink, and S. K. Lodha, "Approaches to uncertainty visualization," *Vis. Compu.*, vol. 13, no. 8, pp. 370–390, 1997.

[70] J. Hullman, P. Resnick, and E. Adar, "Hypothetical outcome plots outperform error bars and violin plots for inferences about reliability of variable ordering," *PLoS One*, vol. 10, no. 11, 2015, Art. no. e0142444.

[71] S. Liu, J. A. Levine, P.-T. Bremer, and V. Pascucci, "Gaussian mixture model based volume visualization," in *Proc. IEEE Symp. Large Data Anal. Visual.*, 2012, pp. 73–77.



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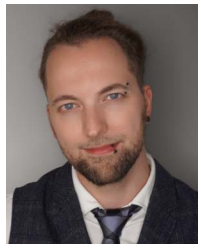
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